

Fig. 4: Actual-size PCB layout of the high-fidelity FM transmitter

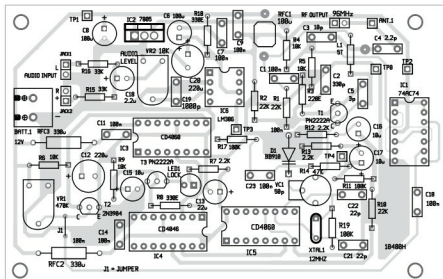


Fig. 5: Components layout for the PCB

and combined in R15 and R16. VR2 is audio-level control. The amplified audio frequency (AF) from IC6

is fed to the varactor diode via R17 and C23 for frequency modulation of the RF signal.

Connect a piece of wire (78cm long) to the RF output as an antenna. You may use an external RF amplifier at RF output to amplify the signal to 1W level.

Construction and testing

An actual-size PCB layout of the high-fidelity FM transmitter is shown in Fig. 4 and its components layout in Fig. 5. House the transmitter in a small plastic box. Keep all leads as short as possible. Use good-quality shielded cable for input audio connections. 12V DC regulated hum-free power supply is recommended.

The operating frequency of the transmitter is 96MHz. Set VR2 to minimum and adjust VC1 for a reading of 3.6V to 4.2V at test point TP4. VR1 is biasing control for transistor T2 and should be set in the middle position. LED1 should light up to indicate a frequency lock at 96MHz. Feed audio signal at JACK1 or JACK2. This signal will be transmitted at 96MHz via antenna ANT.1. Tune the FM receiver to 96MHz. You should be able to hear the transmitted signal in your FM receiver. You can also use your mobile phone if it has inbuilt FM receiver. Vary VR1, VR2 and VC1 for adjustment. For test points and troubleshooting, refer to test points table. **EFY**



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